

Algebraic number theory

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1 Motivation

Let $f(\theta) = \cos(\theta) \cos(2\theta) \cos(4\theta)$. You might already know the following.

Fact 1.1 (Morrie's law). $f(20^\circ) = \frac{1}{8}$.

The proof for this is quite pretty: by three applications of sine double angle, we have

$$\begin{aligned} \cos(20^\circ) \cos(40^\circ) \cos(80^\circ) &= \frac{1}{\sin(20^\circ)} \sin(20^\circ) \cos(20^\circ) \cos(40^\circ) \cos(80^\circ) \\ &= \frac{1}{2} \cdot \frac{1}{\sin(20^\circ)} \sin(40^\circ) \cos(40^\circ) \cos(80^\circ) \\ &= \frac{1}{4} \cdot \frac{1}{\sin(20^\circ)} \sin(80^\circ) \cos(80^\circ) \\ &= \frac{1}{8} \cdot \frac{1}{\sin(20^\circ)} \sin(160^\circ) \\ &= \frac{1}{8}. \end{aligned}$$

Similarly, one can show

$$f(20^\circ) = f(100^\circ) = f(140^\circ) = \frac{1}{8}.$$

Why do 20° , 100° , and 140° give the same values? Of course, the same argument works, but *why*?

We will answer this question by developing some basic algebraic number theory.

2 Theory

2.1 Rings and fields

We'll need some basic abstract algebra. I promise it's useful.

Roughly, a field is a set where you can add, subtract, multiply, and divide. A ring is a set where you can add, subtract, and multiply. For example, $\mathbb{Q}$, $\mathbb{C}$, and $\mathbb{Z}/67\mathbb{Z}$ are fields, and $\mathbb{Z}$ and $\mathbb{Z}/12\mathbb{Z}$ are rings but not fields, since in both rings, you can't divide by 2 — $\frac{1}{2}$ is defined in neither $\mathbb{Z}$ nor $\mathbb{Z}/12\mathbb{Z}$. (However, $\frac{1}{2}$ is defined in $\mathbb{Z}/67\mathbb{Z}$: it's 34, since $2 \cdot 34 \equiv 1 \pmod{67}$.)

If this doesn't make sense, here's the actual definition.

Definition 2.1. A *ring* is a set R along with two binary operations $F^2 \rightarrow F$ *addition* and *multiplication*, written $+$ and $\cdot$, respectively, such that the following hold:

- Addition and multiplication are commutative and associative.
- There exist two elements 0 and 1 such that $a + 0 = a$ and $a \cdot 1 = a$ for all $a \in R$.
- For any $a \in R$, there exists $-a \in R$ such that $a + (-a) = 0$.
- For any $a, b, c \in R$, $a \cdot (b + c) = a \cdot b + a \cdot c$.

We often omit the multiplication symbol, writing ab instead of $a \cdot b$.

A *field* k is a ring where $0 \neq 1$ and for all $a \neq 0$, then there exists $a^{-1} \in k$ such that $a \cdot a^{-1} = 1$.

Some authors say rings don't have to have commutative multiplication, instead calling what we just defined above "commutative rings." These people work with weird things like quaternions.

We can make rings from rings! For a ring R , let $R[x]$ be the smallest ring containing x . For example, $\mathbb{Z}[x]$ is the set of polynomials in x with integer coefficients, and

$$\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}.$$

For a field K , let $K(x)$ be the smallest field containing x . For example, $\mathbb{Q}(x)$ is the set of quotients of polynomials with rational coefficients, and

$$\mathbb{Q}(\sqrt{3}) = \mathbb{Q}[\sqrt{3}] = \{a + b\sqrt{3} \mid a, b \in \mathbb{Q}\}.$$

Exercise 2.2. Verify $\mathbb{Q}(\sqrt{3}) = \mathbb{Q}[\sqrt{3}]$ as above by showing $\mathbb{Q}[\sqrt{3}]$ is a field. You need to show $\frac{1}{\alpha} \in \mathbb{Q}[\sqrt{3}]$ for any $\alpha \in \mathbb{Q}[\sqrt{3}]$.

2.2 Algebraic numbers

Definition 2.3. Some α is an *algebraic number* if it is the root of some polynomial P with rational coefficients. Some α is an *algebraic integer* if it is the root of some monic polynomial P with integer coefficients.

The set of algebraic numbers is denoted $\overline{\mathbb{Q}}$, and the set of algebraic integers is denoted $\overline{\mathbb{Z}}$. It is a fact that $\overline{\mathbb{Q}}$ is a field and $\overline{\mathbb{Z}}$ is a ring.

We will now introduce the incredibly useful concept of a minimal polynomial.

Theorem 2.4 (Existence and uniqueness of minimal polynomial). For an algebraic integer α , there exists a unique polynomial (up to scaling) P_α of minimal degree such that $P_\alpha(\alpha) = 0$.

This P_α is called the *minimal polynomial* of α . The *degree* of α is the degree of P_α . For example, $P_{\sqrt{3}}(x) = x^2 - 3$.

As α is a minimal polynomial, some P_α of minimal degree exists. We need to prove P_α is unique. We will need the following.

Lemma 2.5. Suppose $P(\alpha) = 0$. Then $P_\alpha \mid P$.

Proof. By division algorithm, let $P = qP_\alpha + r$ for polynomials q and r where $\deg r < \deg P_\alpha$. Plugging in α yields $0 = r(\alpha)$. Along with $\deg r < \deg P_\alpha$, this implies r is the zero polynomial. $\square$

The minimal polynomial P_α of an algebraic number α is irreducible, since if $P_\alpha = AB$, where $\deg A, \deg B < \deg \alpha$, $A(\alpha) = 0$ or $B(\alpha) = 0$, contradicting the assumption that P_α is a minimal polynomial.

In the previous section, we saw that by “rationalizing denominators,” $\mathbb{Q}(\sqrt{3}) = \mathbb{Q}[\sqrt{3}]$, since $\mathbb{Q}[\sqrt{3}]$ is a field. In fact, $\mathbb{Q}[\alpha]$ is a field for any algebraic number α . Let me show you why.

I claim that the ring $\mathbb{Q}[\alpha]$ can be thought of as the set $\mathbb{Q}[x]/(P_\alpha(x))$. For example, $\mathbb{Q}[\sqrt{3}]$ can be thought of as the notation $\mathbb{Q}[x]/(x^2 - 3)$. Wait, what does this notation mean? You know how $\mathbb{Z}/(5)$ is the integers modulo 5? In the same way, $\mathbb{Q}[x]/(x^2 - 3)$ is the set of polynomials with coefficients in $\mathbb{Q}$, “modulo $x^2 - 3$.” This means we can add and subtract multiples of $x^2 - 3$ at will. For example, in $\mathbb{Q}[x]/(x^2 - 3)$, we may compute

$$x^3 = x^3 - x(x^2 - 3) = 3x.$$

Why is this the same concept as above? This is best explained with an example. Look at these two parallel calculations: the left is $\mathbb{Q}(\sqrt{3})$, and the right is $\mathbb{Q}[x]/(x^2 - 3)$.

$$\begin{array}{ll} (3 + \sqrt{3})(1 - \sqrt{3}) & (3 + x)(1 - x) \\ = 3 - 2\sqrt{3} - \sqrt{3}^2 & = 3 - 2x - x^2 \\ = 3 - 2\sqrt{3} - \sqrt{3}^2 + (\sqrt{3}^2 - 3) & = 3 - 2x - x^2 + (x^2 - 3) \\ = -2\sqrt{3} & = -2x. \end{array}$$

As you can see, x acts like $\sqrt{3}$. For the experienced reader, the relevant buzzword sequence is that there’s an isomorphism $\mathbb{Q}(\sqrt{3}) \rightarrow \mathbb{Q}[x]/(x^2 - 3)$ given by $1 \mapsto 1$ and $\sqrt{3} \mapsto x$.

Now let me convince you that $\mathbb{Q}(\alpha) = \mathbb{Q}[x]/(P_\alpha(x))$ is a field. Note that $\mathbb{Z}/(5)$ is a field, since 5 is prime. Similarly, $P_\alpha(x)$ is “prime” in $\mathbb{Q}(\alpha)$, as it’s an irreducible polynomial. Therefore, $\mathbb{Q}(\alpha) = \mathbb{Q}[x]/(P_\alpha(x))$ is a field, just like $\mathbb{Z}/(5)$ is a field.

Remark. The above commentary was adapted from my handout “[Demystifying finite fields](#).”

Exercise 2.6. Show that an algebraic number α is an algebraic integer iff P_α has integer coefficients.

The roots of P_α are called the *Galois conjugates* of α . For example, the Galois conjugates $2 + \sqrt{3}i$ are $\{2 + \sqrt{3}i, 2 - \sqrt{3}i\}$. A good general intuition to have is that it’s often good to think of an algebraic number α in terms of its minimal polynomial P_α and its Galois conjugates. More harshly, α is useless by itself. Bring in its brothers and sisters — its Galois conjugates — to fight with it.

3 Cyclotomic polynomials

Definition 3.1. The n th cyclotomic polynomial Φ_n is the monic polynomial with roots as the n th primitive roots of unity.

Immediately, we see

$$\prod_{d|n} \Phi_d(x) = x^n - 1.$$

This implies the following.

Fact 3.2. Cyclotomic polynomials have integer coefficients.

Sketch: if $x^m - 1 = AB$ for polynomials A and B with $A \in \mathbb{Z}[x]$ then $B \in \mathbb{Z}[x]$ as well. Then induct on n to finish.

Funny story about Φ_{105} : people used to think all cyclotomic polynomials had coefficients in $\{-1, 0, 1\}$, but Φ_{105} has a -2 .

Fact 3.3. Cyclotomic polynomials are irreducible.

First, we'll need to invoke the following sledgehammer.

Theorem 3.4 (Dirichlet's theorem). There are infinitely many primes $p \equiv a \pmod{n}$ as long as $\gcd(a, n) = 1$.

Funny one due to Schur: [lehigh](#).

Sick. Now the discriminant of $x^n - 1$ is $\pm n^n$.

Let P be any divisor of $x^n - 1$.

We claim that if ζ is a root, ζ^p is a root as well (as long as $p \nmid n$). This is enough by Dirichlet.

Theorem 3.5 (Theorem 4.8, "Orders Modulo a Prime"). Let p be a prime, n be a positive integer and a any integer. If $p \mid \Phi_n(a)$, then

- a has order n modulo p ,
- $p \mid n$.

Sketch: Suppose $\text{ord}_a(p) = m$ and suppose $m \neq n$. Then as $\Phi_n(x) \mid x^n - 1$,

$$a^n - 1 \equiv 0 \pmod{p},$$

hence $m \mid n$. So a is a double root of $x^n - 1$. Take the derivative to get $na^{n-1} \equiv 0 \pmod{p}$, so $p \mid n$.

TODO: Algebraically irreducible concepts in complex bashing.